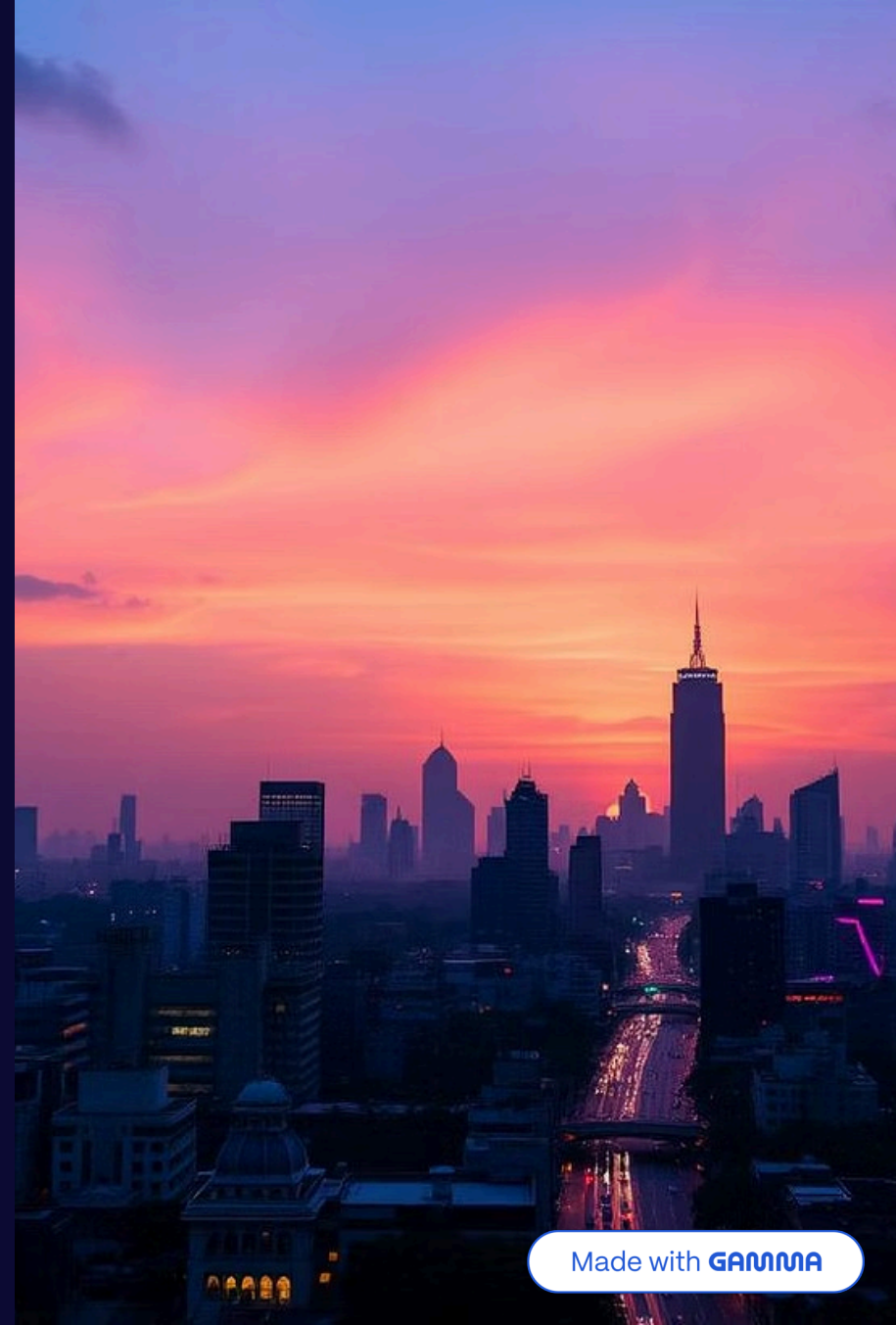


Roommate Compatibility Prediction Using BERT and Ensemble Learning

Presented by Gayatri Kamdar, HSNC University, Mumbai



The Housing and Roommate Challenge in Mumbai

Mumbai faces a severe housing crisis with soaring rents and cramped spaces. Sharing flats is common, but poor roommate matches cause daily stress and conflict, harming residents' well-being. Many have experienced how bad roommates can disrupt life in this bustling city.



Current Solutions and the Research Gap

Platform	Focus
NoBroker	Budget & Location
Roomi	Budget & Location
This Study	Personality & Lifestyle Compatibility

Existing platforms prioritize cost and location, but lack insights into personality or lifestyle compatibility. Our goal is to fill this gap and help people find roommates with whom they truly get along, improving living experiences in Mumbai.

Research Objectives

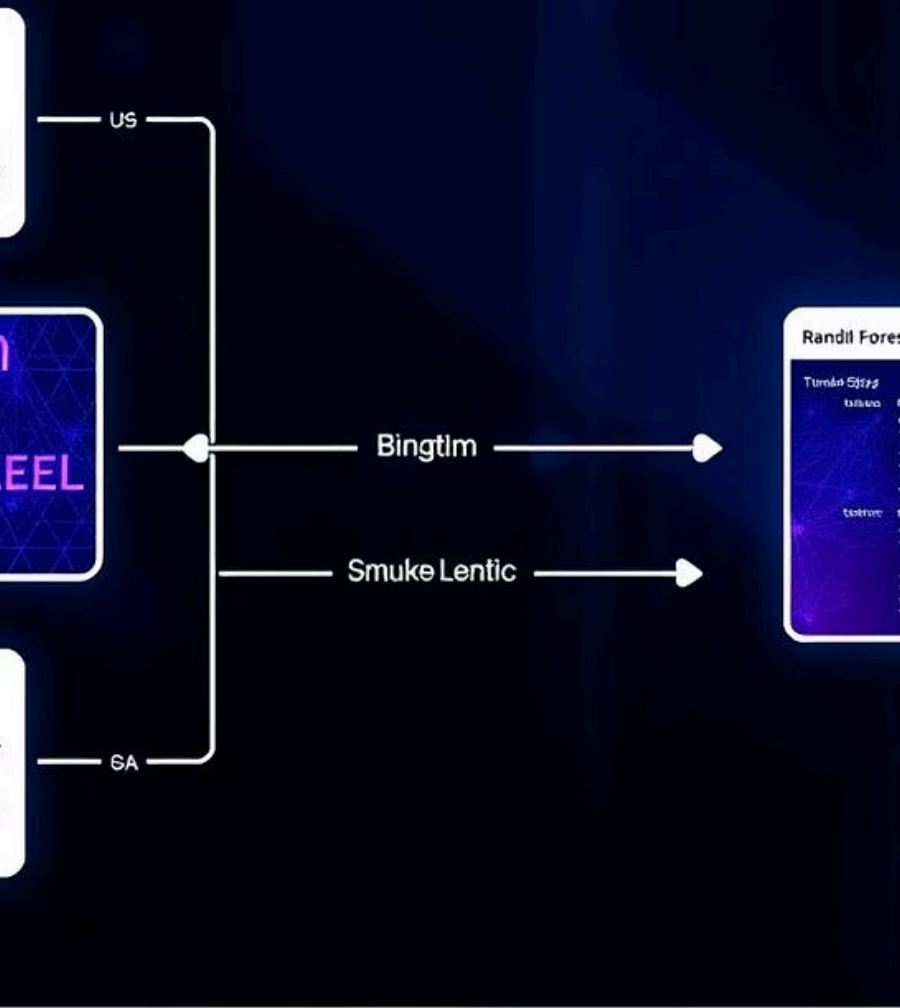
We aim to build a prediction system leveraging both structured data (age, budget) and unstructured lifestyle descriptions. The system is tailored to Mumbai's diverse residents who share tight budgets and living spaces.





Dataset Overview

The dataset contains 1,500 survey responses from Mumbai collected in 2024. Structured features include age, sleep habits, budget, and cleaning frequency. Unstructured data captures lifestyle text such as “I’m quiet, love cooking.” This data reflects real stories from both students and working professionals.



Methodology: A Hybrid Approach

Our approach combines BERT and BiLSTM models to analyze lifestyle text alongside Random Forest algorithms on structured data. Personality Trait Prediction (PTP) infers Big Five traits. The Flask web app offers predictions in about 1.6 seconds, balancing human-like language understanding with robust numerical data processing.



Personality Trait Prediction from Text

Using BERT and BiLSTM, the system infers personality traits such as Agreeableness and Conscientiousness from short lifestyle descriptions. This eliminates the need for time-consuming personality tests. For example, “I’m quiet” predicts a high Conscientiousness score, aiding compatibility matching.

Model Performance and Evaluation

Model	F1-Score
Random Forest	0.85
BERT	0.83
SVM	Lower
Logistic Regression	Lower

The ensemble model achieves an overall accuracy of 85.7% with strong F1-scores, outperforming traditional methods like SVM or Logistic Regression. Combining numerical data and language-based inference nails roommate compatibility predictions effectively.

SHAP Summary

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Key Features Driving Compatibility

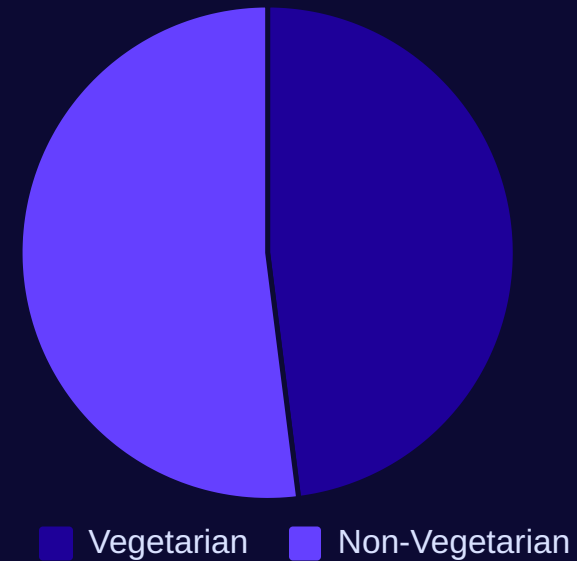
Analysis shows cleaning habits, sleep schedule, Agreeableness scores, and budget are top predictors. SHAP explanations reveal that mismatches in cleanliness or budgeting preferences significantly reduce compatibility, emphasizing practical concerns in Mumbai's shared living situations.

Mumbai-Specific Insights

Budget Sensitivity: Rent matching typically ranges from INR 10,000 to 15,000, reflecting tight financial planning.

Cultural Tolerance: 68% of residents prioritize lifestyle compatibility over religious differences when choosing roommates.

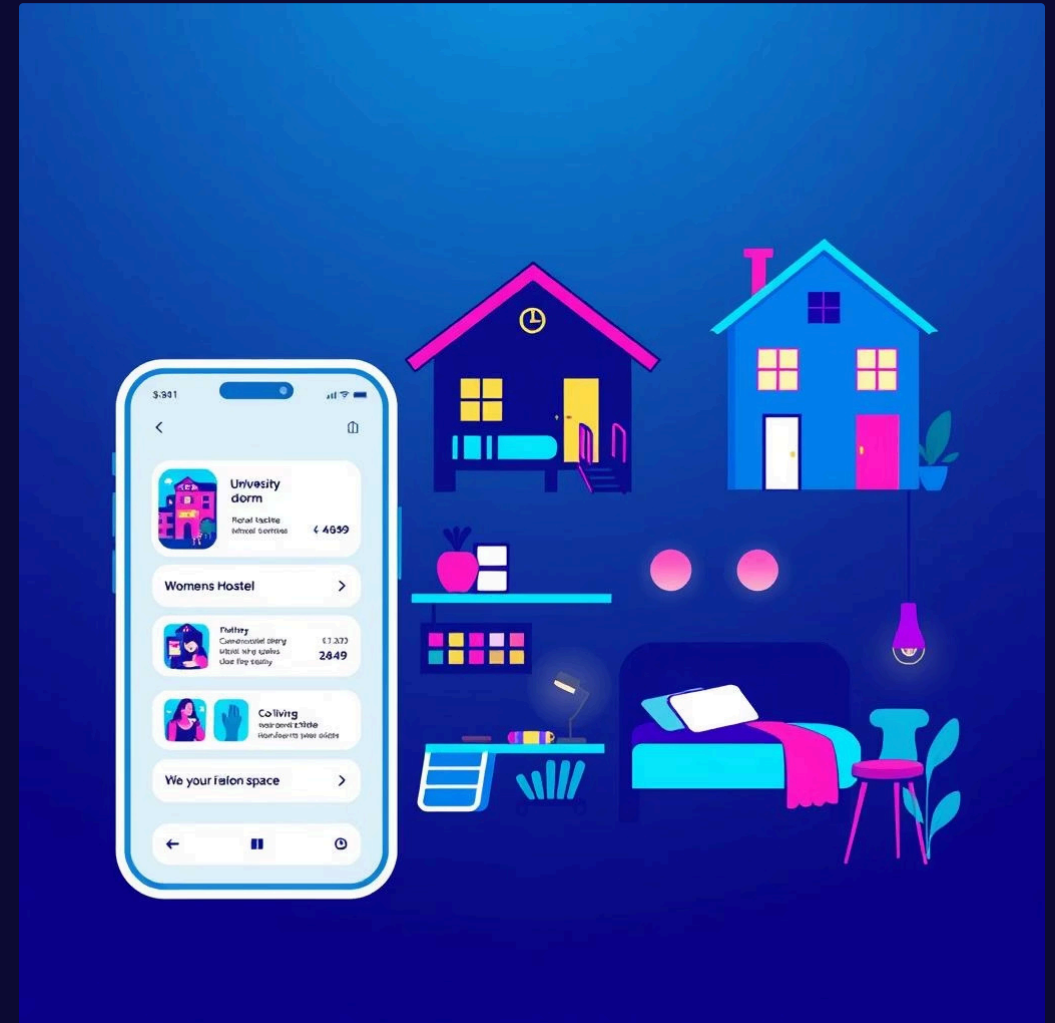
Work-from-Home Impact: With 60% working remotely, noise levels and daily routines play a crucial role in compatibility decisions.



Applications of Roommate Compatibility Prediction

- **Universities:** Enhance dormitory pairings to foster positive living experiences and reduce conflicts.
- **NoBroker:** Introduce a premium feature offering "compatible match" options for renters seeking ideal roommates.
- **Women's Hostels & Co-living Startups:** Improve safety, comfort, and compatibility for residents in shared housing.

"This could make Mumbai homes happier."



Future Work

Integrate User Feedback

Collect and apply real user insights to continually improve matching accuracy and relevance.

Add Voice and Video Inputs

Incorporate multimedia data to enhance personality and compatibility predictions through richer signals.

Expand Dataset Size

Leverage larger, more diverse datasets to strengthen model robustness and generalizability.

Scale to Other Cities

Adapt the model for broader urban contexts beyond Mumbai, addressing unique regional factors.

"I want to make matches even smarter."

Conclusion



High Accuracy

The system achieves 85.7% accuracy in predicting roommate compatibility for Mumbai's shared living environments.



Tech Meets Psychology

Combines advanced BERT language models with psychological insights to improve matchmaking quality and resident satisfaction.



Better Living

Creates less stressful, more harmonious shared homes by helping people find compatible roommates with ease.

Thank You